

Human *RAP2A* Homolog of the *Drosophila* Asymmetric Cell Division Regulator *Rap2l* Targets the Stemness of Glioblastoma Stem Cells

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eLife Assessment

This **useful** study explores the role of RAP2A in asymmetric cell division (ACD) regulation in glioblastoma stem cells (GSCs), drawing parallels to *Drosophila* ACD mechanisms and proposing that an imbalance toward symmetric divisions drives tumor progression. While findings on RAP2A's role in GSC expansion are promising, the study is nevertheless **incomplete**. Limitations include the lack of comprehensive GBM subtype analysis, insufficient mechanistic validation, and reliance on neurosphere models without in vivo confirmation. Addressing those gaps is necessary to substantiate the study's claims.

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Abstract

Asymmetric cell division (ACD) is a fundamental process to balance cell proliferation and differentiation during development and in the adult. Cancer stem cells (CSCs), a very small but highly malignant population within many human tumors, are able to provide differentiated progeny by ACD that contribute to the intratumoral heterogeneity, as well as to proliferate without control by symmetric, self-renewing divisions. Thus, ACD dysregulation in CSCs could trigger cancer progression. Here we consistently find low expression levels of *RAP2A*, the human homolog of the *Drosophila* ACD regulator *Rap2l*, in glioblastoma (GBM) patient samples, and observe that scarce levels of *RAP2A* are associated with poor clinical prognosis in GBM. Additionally, we show that restitution of *RAP2A* in GBM neurosphere cultures increases the ACD of glioblastoma stem cells (GSCs), decreasing their proliferation and expression of stem cell markers. Our results support that ACD failures in GSCs increases their spread, and that ACD amendment could contribute to reduce the expansion of GBM.

Introduction

Asymmetric cell division (ACD) is a universal mechanism for generating cellular diversity during development as well as for regulating tissue homeostasis in the adult (1). Stem and progenitor cells undergo ACD to simultaneously give rise to a self-renewing stem/progenitor cell and to a daughter cell that is committed to enter a differentiation program. Remarkably, over the past years it has become apparent that undermining this critical process can lead to tumor-like overgrowth (2). This link between failures in the process of ACD and tumorigenesis was first established using the *Drosophila* neural stem cells, called neuroblasts (NBs), as a model system (3). In this work, authors showed how *Drosophila* larval mutant brains for ACD regulatory genes were able to induce the formation of massive tumoral masses after weeks of being transplanted into the abdomen of wild-type fly hosts. Intriguingly, some genes firstly identified in *Drosophila* as tumor suppressor genes, such as *discs large 1* (*dlg1*), *lethal* (*l(2)gl*) and *brain tumor* (*brat*), were later uncovered as ACD regulators (4–11), further supporting the connection between compromised ACD and tumorigenesis.

Tumor initiating cells or cancer stem cells (CSCs) were originally identified in acute myeloid leukemia (AML) (12,13) but, to date, several studies have also revealed the existence of CSCs in multiple solid tumors (14). These CSCs represent a very small but highly malignant population within the tumor, able to proliferate without control by symmetrical, self-renewal divisions, as well as to provide differentiated progeny by ACD that contribute to the heterogeneity of the tumor. These properties, along with their quiescent state, increase in metabolic activity and high capacity of DNA repair, make CSCs extremely resistant to radio- and chemotherapy and responsible for malignant relapse (14–18). Thus, understanding the unique nature of these cells and their landmarks is crucial to target CSCs and hence to completely destroy the tumor. An intriguing possibility is that an unbalanced ACD process in CSCs, favoring symmetric renewal divisions, might contribute to the transition from a chronic (low grade) to an acute (high grade) phase in the cancer progression (19,20).

Glioblastoma (GBM), the most aggressive and lethal brain tumor with very poor prognosis, is among the solid tumors in which the presence of CSCs, called glioma or glioblastoma stem cells (GSCs), has been proven. As other CSCs, GSCs are responsible for the formation, maintenance, resistance to conventional therapy and consequent relapse of GBM (21–25). Several GSC biomarkers, such as CD133 and Nestin, and signaling pathways, such as Notch and Hedgehog, have been identified to help isolate and target these crucial cells (14,26,27). Despite of that, there are still many unknowns regarding the nature, properties and origin of CSCs in general and of GSCs in particular. Here, we analyze the consequences of compromising ACD in the GSCs within GBM neurosphere cell cultures. Specifically, we show that *RAP2A*, the human homolog of *Rap2l*, which encodes a *Drosophila* small GTPase and novel ACD regulator, displays low expression levels in GBM, and that restitution of *RAP2A* in GBM neurosphere cultures increases the ACD and decreases the expression of stem cell markers in the GSCs. Our results support that failures in ACD increase GSC proliferation promoting their spreading and that ACD amendment might contribute to reduce the expansion of GBM.

Results

RAP2A is highly downregulated in GBM patients

As a first approach to analyze whether ACD might be impaired in GBM, we analyzed the expression levels of human homologs of 21 *Drosophila* ACD regulators in a GBM microarray (28 [↗](#)) (Fig. 1A [↗](#)). We also investigated which of those genes were more similar in expression pattern to the others by performing a Gene Distance Matrix (Fig. 1B [↗](#)). The human genes *TRIM2* and *RAP2A* showed the lowest levels in the GBM patient samples analyzed. *TRIM2* belongs to the same family that *TRIM3* and *TRIM32*, all of them related to the *Drosophila* ACD regulator gene *brat*. *TRIM3*, the closest homolog of *brat*, has been previously found to be expressed at low levels in GBM (29 [↗](#)). Hence, we concentrated on *RAP2A*, the second human gene consistently found at very low levels in all the patient samples (Fig. 1A [↗](#)). *RAP2A* is also known to have a tumor suppressor role in the context of glioma migration and invasion (30 [↗](#), 31 [↗](#)). First, to further support the microarray data expression levels in a bigger sample size, we performed *in silico* analyses taking advantage of established datasets of Glioblastoma IDH wild-type, such as “The Cancer Genome Atlas” (TCGA) and “Gravendeel cohort”. We confirmed that low *RAP2A* expression levels in human GBM are associated with a poor prognosis (Fig. 1C [↗](#)). Hence, we focused on the human gene *RAP2A* to analyze its potential role in ACD and its relevance in GBM.

Drosophila RAP2A homolog Rap2l regulates ACD

We previously showed that the *Drosophila* gene *Rap1* regulates the ACD process (32 [↗](#)). The closest human homolog of *Drosophila* *Rap1* is *RAP1A*, while human *RAP2A* is more similar to *Drosophila* *Rap2l*, which has not been previously analyzed in the context of ACD. Hence, we first wanted to investigate whether *Rap2l* was, like *Rap1*, implicated in ACD regulation. With that aim, we started analyzing the ACD of neural stem cells (neuroblasts, NBs) and intermediate neural progenitors (INPs) within *Drosophila* larval brain type II NB lineages (NBII) (Fig. 2A [↗](#)) (8 [↗](#), 33 [↗](#), 34 [↗](#)). In NBII lineages, there is one NB that expresses the transcription factor Deadpan (Dpn) and several INPs, which express both Dpn and the transcription factor Asense (Ase) (Fig. 2A [↗](#)). We first observed the presence of ectopic NBs (eNBs; Dpn⁺ Ase⁻) after downregulating *Rap2l* in NBII lineages, while in control lineages only one NB was usually found (Fig. 2B [↗](#)). This result suggested that the process of ACD was impaired within those *Rap2l* mutant NBII lineages. To more directly support that, we analyzed the localization of key ACD regulators in metaphase

NBs and INPs of NBII lineages, such as the apical proteins aPKC and Canoe (Cno), and the cell-fate determinant Numb (35 [↗](#)–37 [↗](#)). In control metaphase NBs and INPs, those ACD regulators form crescents at the apical (aPKC and Cno) or basal (Numb) poles of the dividing cell (Fig. 2C [↗](#)). The apical localization of aPKC was not significantly altered after downregulating *Rap2l* in NBII lineages; however, the localization of Cno and Numb did fail (Fig. 2C [↗](#)). All these results strongly suggested that *Rap2l*, as *Rap1*, was an ACD regulator. Hence, we came back to the human homolog of *Drosophila* *Rap2l*, *RAP2A*, to determine its potential relevance in the context of the ACD of GSCs.

RAP2A expression in GBM neurosphere cultures reduces the stem cell population

After confirming by *in silico* analyses that low *RAP2A* expression levels in GBM patients were associated to a poor prognosis, we decided to use established GBM cell lines to investigate in neurosphere cell cultures whether the downregulation of *RAP2A* was affecting the properties of GSCs. We tested six different GBM cell lines finding similar mRNA levels of *RAP2A* in all of them, and significantly lower levels than in control Astros (Fig. 3A [↗](#)). We focused on the GBM cell line called GB5, which grew well in neurosphere cell culture conditions, for further analyses. Given that stem cell markers, such as CD133, SOX2 and NESTIN are landmarks of GSCs, we started analyzing their expression after restoring *RAP2A* in the neurosphere culture (Fig. 3B [↗](#)). Immunofluorescences of each of those markers revealed a significant reduction in the protein intensity in the *RAP2A*-expressing neurospheres (GB5-*RAP2A*) compared to the control

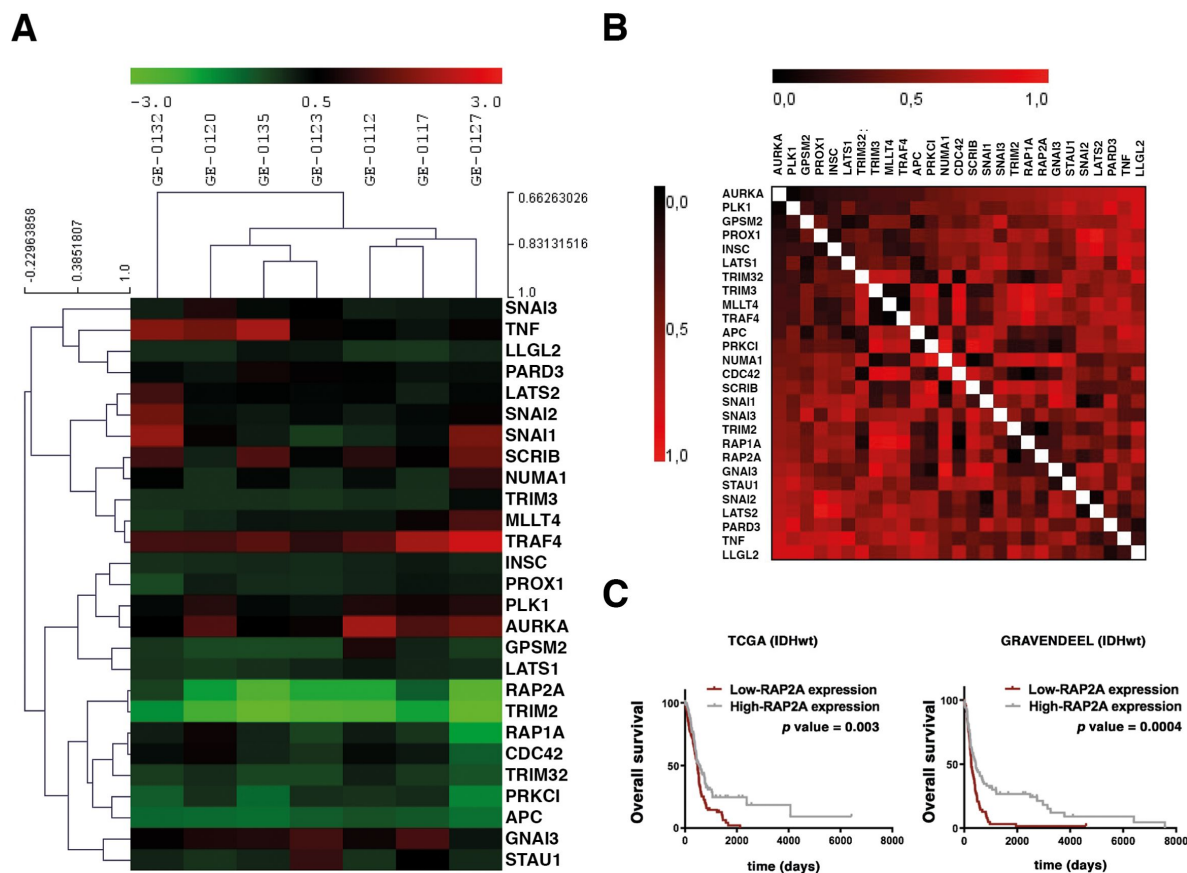


Figure 1.

RAP2A is highly downregulated in GBM patients.

(A, B) Analysis of GBM patient versus control samples focused on the level of expression of human homologs of *Drosophila* ACD regulators by Hierarchical clustering

(A) and the Gene Distance Matrix (B). Color-coded scale bar indicates fold level of expression. (C) Kaplan-Meier survival curves corresponding to patients in the TCGA (IDHwt) and Gravendeel (IDHwt) cohort in GBM datasets. Low *RAP2A* expression levels in human GBM are associated with a poor prognosis.

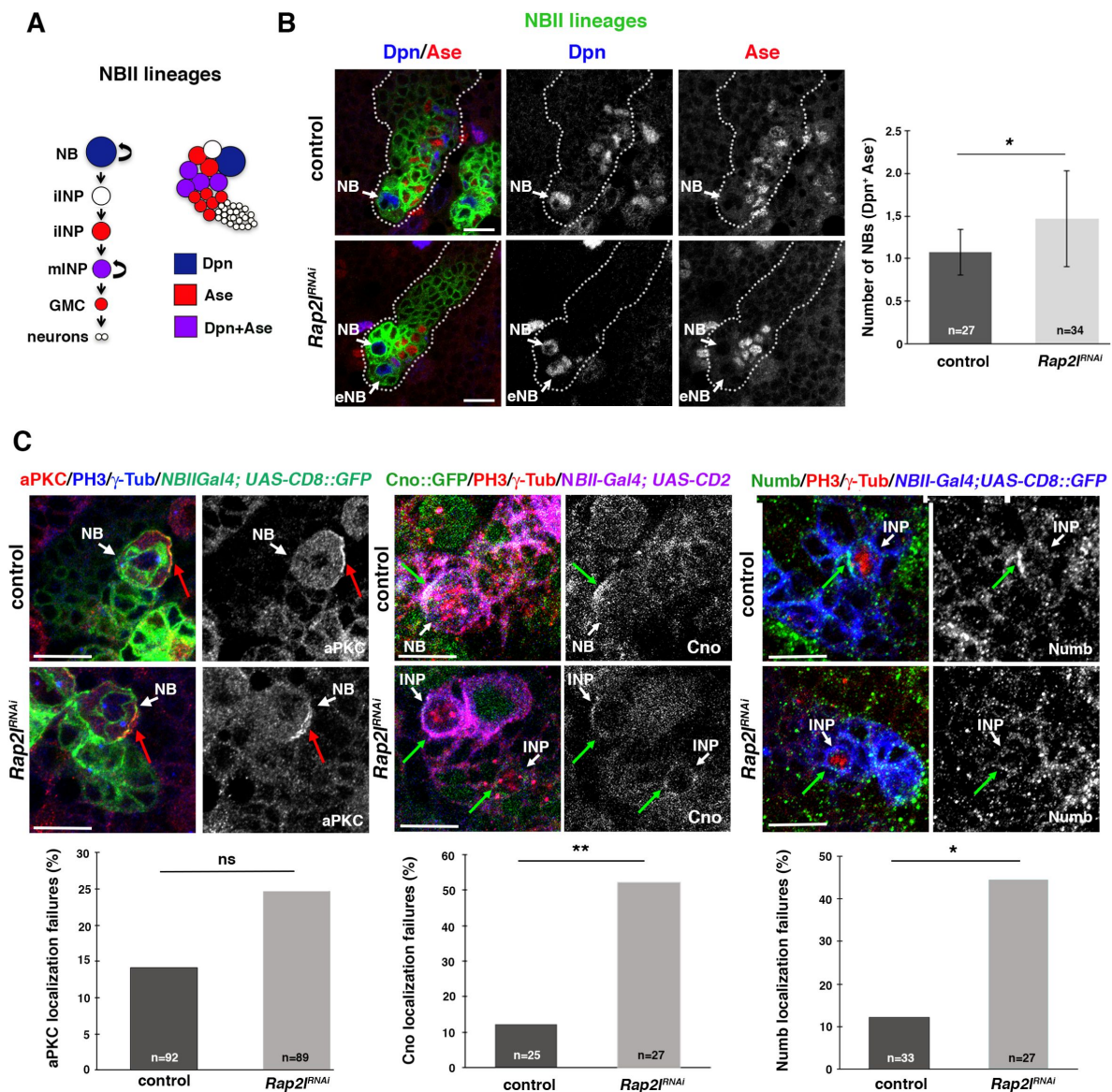


Figure 2.

Drosophila RAP2A homolog *Rap2l* regulates ACD.

(A) *Drosophila* NBII lineage; the only NB per lineage expresses the transcription factor Dpn, while mature INPs express both Dpn and Ase; iINP, immature INP; mINP, mature INP; GMC, Ganglion Mother Cell. **(B)** Confocal micrographs showing a control larval brain NBII lineage with one NB (Dpn⁺ Ase⁺) and an NBII lineage in which *Rap2l* has been downregulated displaying an ectopic NB (eNB). **(C)** *Rap2l* downregulation in NBII lineages by the *wor-Gal4 ase-Gal80* (an NBII-specific driver) causes Cno and Numb localization failures (green arrows) in dividing progenitors (NB or INPs), while aPKC localization is not significantly altered (red arrows). Data shown in the scaled bar graphs was analyzed with a Chi-square test, * $p < 0.05$, ** $p < 0.01$, ns, not significant, n=number of NB lineages analyzed (in B) or number of dividing cells analyzed (in C); scale bar: 10 μ m

neurospheres (GB5) (**Fig. 3C** [↗](#)). Likewise, RT-PCRs and Westerns blots also showed a significant decrease in the mRNA and protein expression levels, respectively, of those stem cell markers (**Fig. 4** [↗](#)). Hence, RAP2A seems to be contributing to decrease the stem cell population within GBM neurosphere cultures.

RAP2A expression in GBM neurosphere cultures decreases cell proliferation and sphere size

Next, we wondered whether the reduction in the stem cell population after expressing *RAP2A* in neurosphere cultures was reflected in the level of cell proliferation and, consequently, in the size of the neurospheres. To investigate this, we first analyzed the expression of the proliferation marker and key tool in cancer diagnostics Ki-67, highly expressed in cycling cells but absent in quiescent (G0) cells ([38](#) [↗](#), [39](#) [↗](#)). We detected a significant decrease in the number of proliferating cells per neurosphere in those cultures expressing *RAP2A* (**Fig. 5A** [↗](#)). Moreover, the average area size of these GB5- *RAP2A* neurospheres was also significantly reduced compared to control GB5 neurospheres (**Fig. 5B** [↗](#)). Thus, *RAP2A* promotes a decrease in cell proliferation.

RAP2A expression in GBM neurosphere cultures fosters ACD in GSCs

Given the reduction in cell proliferation and in the stem cell population after adding *RAP2A* to the control GB5 neurosphere cultures, we wanted to investigate whether *RAP2A* was favoring an asymmetric mode of cell division in these cells. Thus, we decided to follow over time the GB5-*RAP2A* and the control GB5 neurospheres, analyzing the number of cells per neurosphere cluster. We reasoned that an even number of cells per cluster would be indicative of symmetric cell divisions, while an odd number of cells would point to an asymmetric mode of cell division. We observed a significant increase of cell clusters with an odd number of cells in the GB5-*RAP2A* neurospheres (n= 124 cell clusters analyzed) compared to control GB5 neurospheres (n=153 cell clusters) (**Fig. 6A** [↗](#)). This result was very suggestive of an increase in the number of ACDs in GB5-*RAP2A* neurosphere cells. To further support this result, we decided to analyze the distribution of the cell fate determinant NUMB, a key conserved ACD regulator whose asymmetric distribution in the mother and progeny ultimately promotes the repression of stem cell self-renewal in the daughter cell in which is segregated ([40](#) [↗](#)–[44](#) [↗](#)). We observed a significant increase in the number of GB5- *RAP2A* dividing cells that showed an asymmetric localization of NUMB in the progeny, compared to control GB5 dividing cells (**Fig. 6B** [↗](#)). Thus, all these results strongly supported a function of *RAP2A* promoting ACD in the GSCs.

Discussion

ACD is an evolutionary conserved mechanism used by stem and progenitor cells to generate cell diversity during development and to regulate tissue homeostasis in the adult. CSCs, present in many human tumors, can divide asymmetrically to generate intratumoral heterogeneity or symmetrically to expand the tumor by self-renewal. Over the past 15 years, it has been suggested that dysregulation of the balance between symmetric and asymmetric cell divisions in CSCs, favoring symmetric divisions, can trigger tumor progression in different types of cancer ([19](#) [↗](#), [20](#) [↗](#), [45](#) [↗](#)) including mammary tumors ([43](#) [↗](#), [46](#) [↗](#)), glioblastoma ([29](#) [↗](#)), oligodendrogliomas ([47](#) [↗](#), [48](#) [↗](#)), colorectal cancer ([49](#) [↗](#), [50](#) [↗](#)) and hepatocellular carcinoma ([50](#) [↗](#)). Thus, it is of great relevance to get a deeper insight into the network of regulators that control ACD, as well as the mechanisms by which they operate in this key process.

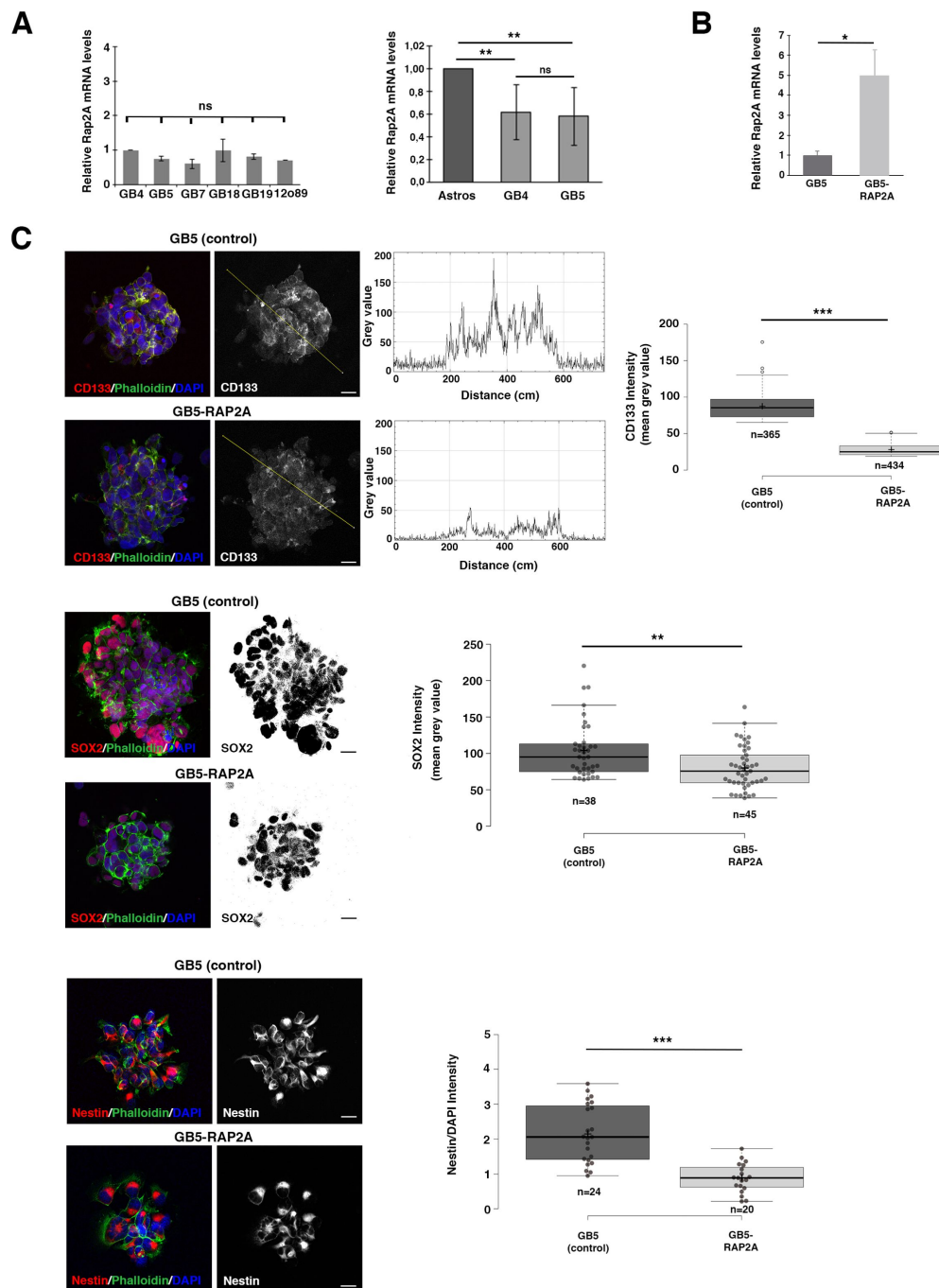


Figure 3.

RAP2A expression in GBM neurosphere cultures reduces the stem cell population.

(A) Different GBM cell lines show similar *RAP2A* mRNA levels and significantly lower levels than in control Astrocytes. **(B)** *RAP2A* mRNA levels are significantly higher in the GB5 line after infecting this line with *RAP2A* (GB5-RAP2A). Data shown in the scaled bar graphs in A and B was analyzed with an ANOVA and a T-test, respectively; error bars show the SD; $n = 2$ (in A) and 3 (in B) different experiments. **(C)** Immunofluorescences of the GSC stem cell markers CD133, SOX2 and Nestin reveal a significant reduction in the protein intensity in the GB5 *RAP2A*-expressing neurospheres (GB5-RAP2A) compared to the control neurospheres (GB5). Data shown in the box plots was analyzed with a U Mann-Whitney for CD133 and Nestin and with a T-test for Sox; the central lines represent the median and the box limits the lower and upper quartiles, as determined by R software; crosses represent sample means; error bars indicate the SEM; n = number of sample points; p values: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns, not significant; scale bar: 10 μ m.

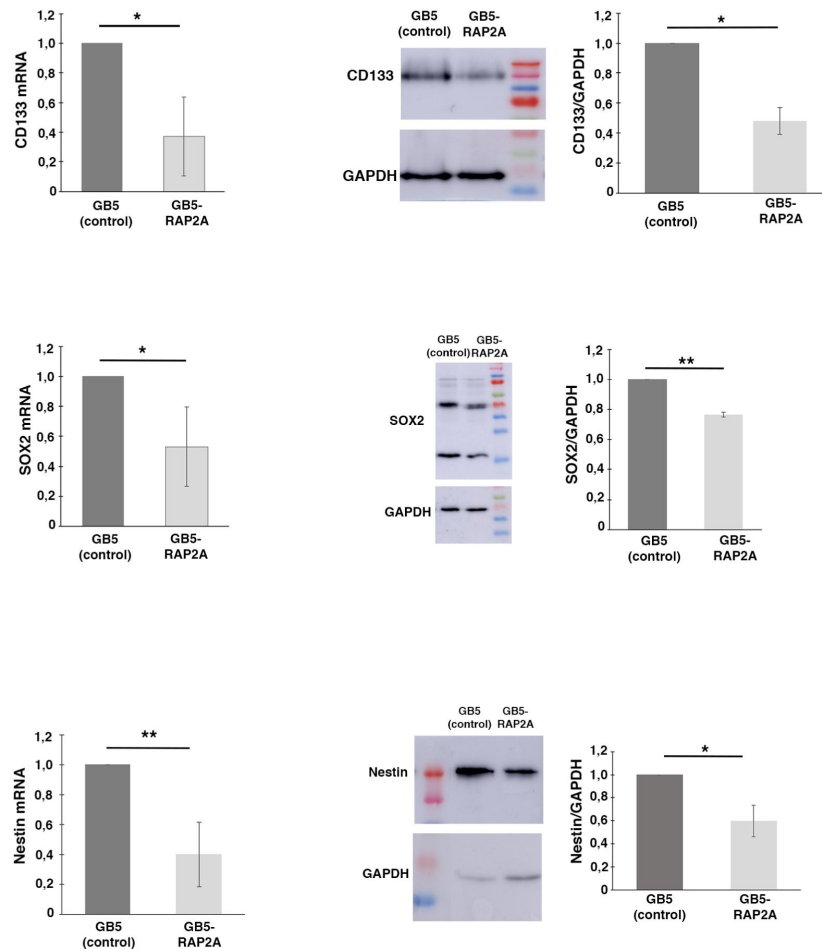
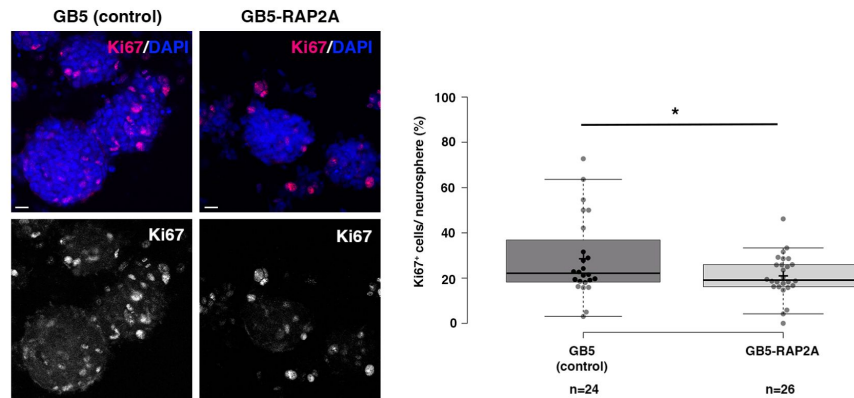


Figure 4.

RAP2A expression in GBM neurosphere cultures reduces the stem cell population.

RT-PCRs and Westerns blots show a significant decrease in the mRNA and protein expression levels, respectively, of the GSC markers CD133, SOX2 and Nestin in the GB5 RAP2A-expressing neurospheres (GB5-RAP2A) compared to the control neurospheres (GB5). Data shown in the scaled bar graphs was analyzed with an unpaired two-tailed Student's *t* test; error bars show the SD; *n* = 3 different experiments; *p* values: **p* < 0.05, ***p* < 0.01.

A



B

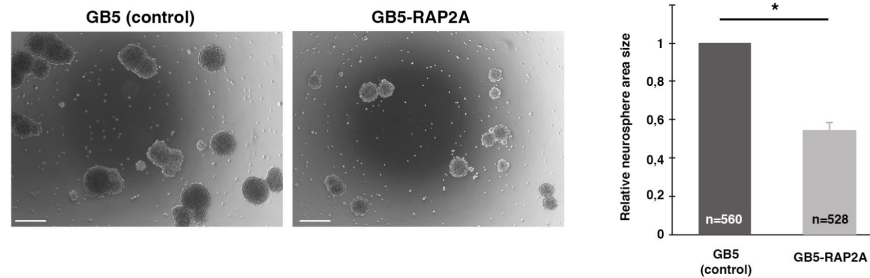


Figure 5.

RAP2A expression in GBM neurosphere cultures decreases cell proliferation and sphere size.

(A) GB5 neurospheres expressing RAP2A (GB5-RAP2A) show a significantly lower number of Ki67 expressing cells per neurosphere than control GB5 neurospheres. Data shown in the box plots was analyzed with a T-test; the central lines represent the median and the box limits the lower and upper quartiles, as determined by R software; crosses represent sample means; error bars indicate the SEM; n= number of sample points. **(B)** GB5 neurospheres expressing RAP2A (GB5-RAP2A) show a significant decrease in their size compared to control GB5 neurospheres of the same stage. Data shown in the scaled bar graphs was analyzed with an unpaired two-tailed Student's T- test; error bars show the SD; n= total number of neurospheres of 3 different experiments; *p* values: **p*<0.05; scale bars: 10 μ m (in A) and 100 μ m (in B).

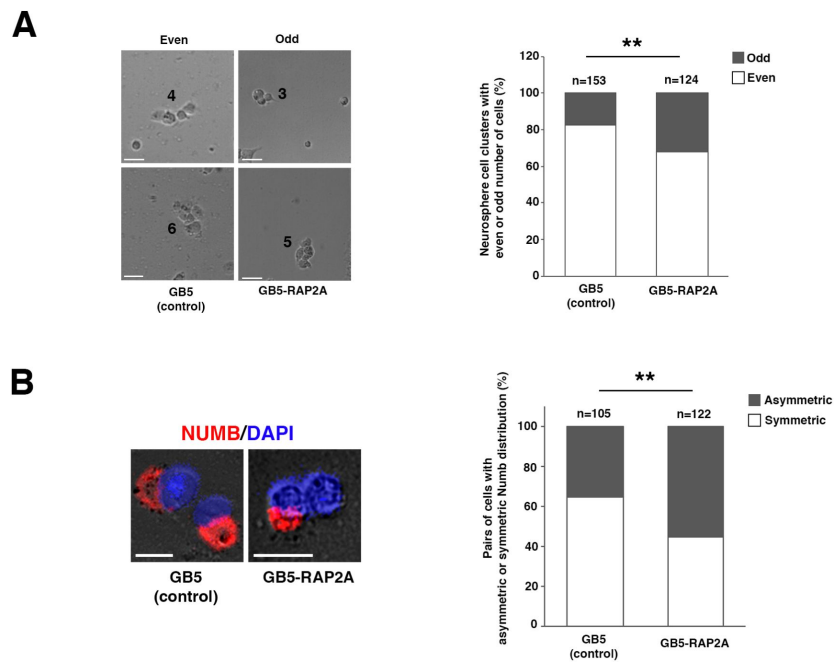


Figure 6.

***RAP2A* expression in GBM neurosphere cultures fosters ACD in GSCs.**

(A) Early stage GB5 neurospheres expressing *RAP2A* (GB5-*RAP2A*) show an odd number of cells significantly more frequently than control GB5 neurospheres, which show more frequently an even number of cells. **(B)** GB5-*RAP2A* dividing cells show a significant increase in the number of asymmetric NUMB localization in the progeny, compared to control GB5 dividing cells. Data shown in the bar graphs was analyzed with a Chi-Square test with Yates correction; n= number of neurosphere cell clusters analyzed. *p* values: ***p*<0.01; scale bar: 20 μ m.

Drosophila neural stem cells or NBs have been used as a paradigm for many decades to study ACD (51). NBs divide asymmetrically to give rise to another self-renewing NB and a daughter cell that will start a differentiation process. Over all these years, a complex network of ACD regulators that tightly modulate this process has been characterized. For example, the so-called cell-fate determinants, including the Notch inhibitor Numb, accumulate asymmetrically at the basal pole of mitotic NBs and are exclusively segregated to one daughter cell, promoting in this cell a differentiation process. The asymmetric distribution of cell-fate determinants in the NB is, in turn, regulated by an intricate group of proteins asymmetrically located at the apical pole of mitotic NBs, generically known as the “apical complex”. This apical complex includes kinases (i.e. aPKC), small GTPases (i.e. Cdc42, Rap1) and Par proteins (i.e. Par-6, Par3), among others (51).

Given the potential relevance of ACD in CSCs, we decided to take advantage of all the knowledge accumulated in *Drosophila* about the network of modulators that control asymmetric NB division. As a first approach, we aimed to analyze whether the levels of human homologs of known *Drosophila* ACD regulators were altered in human tumors. Specifically, we centered on human GBM, as the presence of CSCs (GSCs) have been shown in this tumor. The microarray we interrogated with GBM patient samples had some limitations. For example, not all the human genes homologs of the *Drosophila* ACD regulators were present (i.e. the human homologs of the determinant Numb). Likewise, we only tested seven different GBM patient samples. Nevertheless, the output from this analysis was enough to determine that most of the human genes tested in the array presented altered levels of expression. We selected for further analyses *RAP2A*, one of the human genes that showed the lowest levels of expression, compared to the control samples. However, it would be interesting to analyze in the future the potential consequences that altered levels of expression of the other human homologs in the array can have in the behavior of the GSCs. In silico analyses, taking advantage of the existence of established datasets, such as the TCGA, can help to more robustly assess, in a bigger sample size, the relevance of those human genes expression levels in GBM progression, as we observed for the gene *RAP2A*.

We have previously shown that *Drosophila* Rap1 acts as a novel NB ACD regulator in a complex with the apical regulators Cno (Cno), aPKC and Par-6 (32). Here we have shown that *Drosophila* Rap2l also regulates NB ACD by ensuring the correct localization of the ACD modulators Cno and Numb. We also have demonstrated that *RAP2A*, the human homolog of *Drosophila* Rap2l, behaves as an ACD regulator in GBM neurosphere cultures, and its restitution to these GBM cultures, in which is present at low levels, targets the stemness of GSCs increasing the number of ACDs. Thus, this study supports the relevance of ACD in CSCs of human tumors to refrain the expansion of the tumor. Other studies, however, claim that ACD should be targeted in human tumors as it promotes the intratumoral heterogeneity that hampers the complete tumor loss after chemotherapy (52–54). More studies should be carried out with other human genes homologs of ACD regulators to further confirm the results of this study. Likewise, analyses in vivo (i.e. in mouse xenografts) would be also required to reinforce our conclusions. This would be very relevant in order to consider ACD restitution in CSCs of human tumors, what has been called “differentiation therapy” (55), as a potential alternative therapeutic treatment.

Materials and Methods

Drosophila husbandry, strains and genetics

The following fly stocks were used: *wor-Gal4 ase-Gal80* (56); *UAS-CD8::GFP* (Bloomington *Drosophila* Stock Center -BDSC- #5137); *UAS-CD2* (BDSC #1373); *UAS- Rap2l^{KKRNAi}* (Vienna *Drosophila* Resource Center #107745); *cno::GFP* (57). All the fly stocks were raised and kept in 18°C or 25°C incubators. Experimental temperatures for the assays were maintained using 25°C or 29°C incubators. The *Gal-4* x *UAS*-crosses were carried out at 29°C (the first 2 days they were kept at 25°C, and then transferred to a new tube and left at 29°C) until third instar larvae developed.

***Drosophila* histology, immunofluorescence and microscopy**

Larval brains were dissected out in PBS and fixed with 4% PFA in PBT (PBS and Triton X- 100 0.1%) for 20 min at room temperature with gentle rocking. Fixed brains were washed 3 times for 15 min with PBT (PBS and Triton X-100 0.3%) and then incubated in PBT-BSA for at least 1h before incubation with the corresponding primary antibody/antibodies. The following primary antibodies were used in this study: guinea pig anti-Dpn (1:2,000) ([57](#)), rabbit anti-Ase (1:100) ([57](#)), rabbit anti PKC ζ (1:100; Santa Cruz Biotechnology, sc-216), goat anti-Numb (1:200; Santa Cruz Biotechnology, sc- 23579), mouse anti-PH3 (1:2,000; Millipore, 05-806), mouse anti- γ Tub (1:200; Sigma- Aldrich, T5326) and mouse anti-CD2 (1:50; BioRad). Secondary antibodies conjugated to fluorescent dyes (Invitrogen) were used at 1:400 dilution. Samples were mounted in VECTASHIELD antifade mounting medium for fluorescence (Vector Laboratories, H- 1000). Fluorescent images were captured using a Super-resolution Inverted Confocal Microscope Zeiss LSM 880-Airyscan Elyra PS.1. Images were analyzed using the image processing package FIJI from ImageJ and assembled using Adobe Photoshop CS6.

GBM microarray

The cohort of patients consisted of seven GBM biopsies obtained from the Valencian Network of Biobanks. The procedures for obtaining tissue samples were developed in accordance with national ethical and legal standards, and following the guidelines established in the Declaration of Helsinki, and approved of Clinical Research Ethics Committee of the Hospital General Universitario de Elche, Spain. All patients had been diagnosed with GBM grade IV, and all of them exhibited an *IDH1* plus *IDH2* wild-type genotype. Normal human brain RNA (frontal lobe; pool from 5 donors) was purchased from BioChain and used as a control for gene expression experiments. A total of 12,516 genes were used to generate a custom Agilent Two-Color 8 x 15 K Agilent gene expression DNA microarray, which were deposited in the NCBI Gene Expression Omnibus (GEO) Database under accession number GSE182697 ([28](#)).

Bioinformatics analysis

The level of expression of human homologs of *Drosophila* ACD regulators was assessed in seven different GBM patients versus control samples of the custom-made microarray (GSE182697) ([28](#)). Statistical analysis of microarray data was done in Multiexperiment Viewer (MeV) ([58](#)). Hierarchical clustering analyses were done optimizing gene and sample leaf order and using Pearson Correlation for the distance metric selection. Average linkage clustering was used as a linkage method. For the Gene Distance Matrix (GDM), the distance, inverse of similarity, between two genes was calculated using a distance metric as indicated in MeV.

Human samples and cell lines

Glioma samples were obtained with patients' written informed consent, in accordance with the Declaration of Helsinki, and approved by the Ethics Committee at Hospital 12 de Octubre, Madrid, Spain (CEI 14/023). The primary glioblastoma cell lines were provided by the Biobank at "Hospital 12 de Octubre". Fresh tissue samples were enzymatically dissociated using Accumax (Millipore) and cultured in stem cell medium (Neurobasal, Invitrogen), supplemented with N2 (1:100, Invitrogen), GlutaMAX (1:100, Invitrogen), penicillin-streptomycin (1:100, Lonza), 0.4% heparin (Sigma-Aldrich), 40 ng/ml EGF and 20 ng/ml bFGF2 (Peprotech). The GB cell lines used in this study were: GB4, GB5, GB7, GB18, GB19, 12o89, Astros ([59](#), [60](#)) and HEK293T. The cell lines HEK293T and Astros were maintained in Dulbeccó's modified Eagle Medium supplemented with 10% fetal bovine serum 2 mM L-glutamine, 0.1% penicillin (100 U/ml) and streptomycin (100 μ g/ml) and Non-Essential Aminoacids (NEA 1x).

GBM-derived neurosphere cell cultures

For stem cell culture of GBM-derived neurospheres we used DMEM:F12 with FGF-2 (20 ng/ml), EGF-2 (20 ng/ml), and N2 supplements, 0.1% penicillin (100 U/ml) and streptomycin (100 µg/ml), as previously described (61,62). The tumor spheres were dissociated to single cells using Accumax, and 1×10^5 dissociated cells/ml were plated in a Flask and cultured for 6-7 days.

In Silico Analysis

Data from The Cancer Genome Atlas (TCGA) and the Chinese Glioma Genome Atlas (CGGA) for GBM cohorts were accessed through UCSC Xena-Browser (<https://xenabrowser.net>), and Gliovis (<http://gliovis.bioinfo.cnio.es>) for extraction of overall survival data, gene expression levels, and distribution of genetic alterations. Kaplan-Meier survival curves were generated by stratifying samples into low- and high- expression groups for each gene. Survival differences between these groups were assessed using the log-rank test.

DNA Constructs and Lentiviral Production

Lentiviral vectors were used to produce cells overexpressing *RAP2A* (pLJM1RAP2A #Plasmid 19311, Addgene) or eGFP (pLJMEGFP #Plasmid 19319, Addgene) as infection control. Infected cells were selected with Puromycin. To obtain the lentivirus, HEK293T cells were transiently co-transfected with 5 µg of the appropriate lentiviral plasmid (pLJEGFP or pLJRRap2A, respectively), 5 µg of the packaging plasmid pCMVdr8.74 (#Plasmid 22036, Addgene), and 5 µg of VERSUSV-G envelope protein plasmid pMD2G (#Plasmid 12259, Addgene), using Lipofectamine Plus reagent (Invitrogen, Carlsbad, CA, USA). Lentiviral supernatants were collected after 48 h of HEK293T transfection.

Neurosphere Immunofluorescences and Imaging

For immunostaining, floating cultured neurospheres were incubated on Matrigel coated micro slide glass for 2 hours at 37°C. After having been washed, the attached neurospheres were fixed in 4% paraformaldehyde for 20 min. Cells were blocked for 1 h in 1% FBS and 0.1% Triton X-100 in PBS; then, they were incubated for 2 hours with the primary antibody: mouse anti-CD133 (HB#7, Developmental Studies Hybridoma Bank), mouse anti-Ki67 (3E6, Developmental Studies Hybridoma Bank), rabbit anti- Sox2 (N1C3, GTX101507 GeneTex), mouse anti-Nestin (10c2, sc-23927, Santa Cruz Biotechnology) and mouse anti-Numb (48, sc-136554, Santa Cruz Biotechnology). The secondary antibodies used were Alexa555 goat anti-rabbit (A32732, Invitrogen) or Alexa555 goat anti-mouse (A32727, Invitrogen). Phalloidin 633 (Sigma-Aldrich, 68825 Phalloidin-Atto 633) was added and incubated for 20 min in PBS; Vectashield Mounting medium with DAPI (Linaris, H-1200) was added before mounting. Fluorescent images were captured using a Super-resolution Inverted Confocal Microscope Zeiss LSM 880- Airyscan Elyra PS.1. Images were analyzed using the image processing package FIJI from ImageJ and assembled using Adobe Photoshop CS6 program.

Immunofluorescent quantification

Fluorescence intensity measurements were carried out using ImageJ software package. In all cases, images of neurospheres transfected with GFP (GB5, control) or RAP2A (GB5-RAP2A) lentivirus were analyzed:

CD133: Using the “line tool”, a line was drawn on the channel with the CD133 signal on three different pictures per genotype. Fluorescence was measured across the line sections with the “Plot Profile Tool”, and reported as a subset of images showing the signal intensity plot. For quantification, only the 100 highest gray value points, were taken.

Sox: After adjusting the threshold on the channel with the Sox2 signal and eliminating particles smaller than 5 microns, the fluorescence intensity was measured as the mean gray value (sum of all pixels with gray values inside the selection, divided by the number of pixels). In **Figure 3** [3](#) is shown one example out of three quantifications achieved with at least 20 cells per genotype.

Nestin: The regions of interest (ROIs) were set with the “hand drawing tool” using the Phalloidin staining to follow the cell periphery. At least 20 cells (ROIs) from three different pictures were selected per condition. The background fluorescence was subtracted from the mean gray value fluorescence for each well, and the fluorescence intensity was normalized by the nuclear DAPI signal coming from the same cell (ROI). Plot in **Figure 3C** [3](#) represents the mean gray value that is the sum of all pixels with gray values inside the selection divided by the number of pixels.

RNA isolation and qRT-PCR assay

Total RNA was extracted from 7 days old neurospheres using TRI reagent (AM9738, Invitrogen) following manufacturer indications and quantified using a nanodrop (ND- 1000, Thermo Scientific). 2ug of RNA was treated with DNaseI (EN0521, Thermo Scientific) and reverse transcribed with SuperScript™ III Reverse Transcriptase (1808044, Invitrogen). Quantitative real time PCR (qRT-PCR) was performed using Power SYBR Green PCR Master Mix (PN4367218, Applied Biosystems), following established protocols with 60°C for annealing/extension and 40 cycles of amplification on a QuantStudio™ 3 apparatus (applied Biosystems). Act5 primers were used for normalization, and relative expression was calculated using the comparative $\Delta\Delta C_t$. The average of at least 3 independent experiments is shown in **Figure 4** [4](#).

Neurosphere Western Blots

Protein extracts were prepared by re-suspending cell pellets in a lysis buffer (50-mM Tris, pH 7.5, 300-mM NaCl, 0.5% SDS, and 1% Triton X-100) and incubating the cells for 10 min at 95°C. The lysed extracts were centrifuged at 13,000 g for 10 min at room temperature, and the protein concentration was determined using a commercially available colorimetric assay (BCA Protein Assay Kit). Approximately 25 μ g of protein was resolved by 10% or 12.5% SDS-PAGE, and they were then transferred to a PVDF membrane (Hybond-ECL, Amersham Biosciences). The membranes were blocked for 1 h at room temperature in TBS-T (10-mM Tris-HCl, pH 7.5, 100-mM NaCl, and 0.1% Tween-20) with 5% skimmed milk, and then incubated overnight at 4°C with the corresponding primary antibody diluted in TBS-T with 2.5% skimmed milk. After washing 3 times with TBS-T, the membranes were incubated for 2 h at room temperature with their corresponding secondary antibody, diluted in TBS-T with 2.5% skimmed milk. The proteins were visible by enhanced chemiluminescence with ECL (Pierce) using Amersham imager 680, and the signal was quantified by Fiji-ImageJ software.

Proliferative index (Ki67 analysis)

The number of Ki67+ cells with respect to the total number of nuclei labelled with DAPI within a given neurosphere were counted to calculate the Proliferative Index (PI), which was expressed as the % of Ki67+ cells over total DAPI+ cells. More than 20 neurospheres coming from different pictures were analyzed.

Sphere size measurement

Glioblastoma cells (GB5 cell line) were infected by control lentivirus (LV-GFP) or by lentivirus directing expression of *RAP2A* (LV-*RAP2A*). The tumor spheres were Accumax-dissociated to single cells, during three consecutive passages. Seven days after the third plating in a 6-well plate, spheres were captured using a 10x objective (Carl Zeiss Axio Imager A1 microscope). 20 random pictures of each genotype were taken and the images were analyzed by measuring the area of all the spheres present, using the “freeform drawing tool” in the ImageJ software package. The data presented show the average area measured in 3 independent experiments.

Pair assay and Numb segregation analysis

Dissociated cells were plated at low density on Matrigel-coated glass slides. Cultures were incubated for 21 h at 37°C with 5% CO₂, fixed and immunolabeled for Numb, as well as for DAPI to identify nuclei. First, isolated groups of cells (most probably coming from the same single progenitor) were identified and the number of cells in each cluster was annotated. For each identified DAPI-labeled pair, the localization of Numb was visualized and scored from at least 100 cell pairs. Images in **Figure 6** [↗](#) were acquired in an Inverted Microscope Leica Thunder Imager.

Statistical analysis

The data were first analyzed using the Shapiro-Wilk test to determine whether the sample followed a normal distribution. Parametric T-test or a nonparametric two-tailed Mann-Whitney U test for those that did not follow a normal distribution were used to compare statistical differences between two different groups. Mean and standard deviation values were calculated by standard methods. A Chi-square test was used to analyze the data in **Figure 2C** [↗](#). To assume statistical significance, *p*-values were determined below 0.05. Sample size (*n*) and the *p*-value are indicated in the figure or figure legend; * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001, ns: not significant (*p* > 0.05).

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Additional information

Author contributions

M.F. participated in the design of the experiments, conducted most of them and analyzed the data; R.G. developed the GB lines and performed the *in silico* analyses; V.M.B. and M.S. carried out the GB microarray and bioinformatics analysis; A.C. conceived the study and wrote the manuscript.

References

1. Knoblich JA (2008) **Mechanisms of asymmetric stem cell division** *Cell* **132**:583–97
2. Knoblich JA (2010) **Asymmetric cell division: recent developments and their implications for tumour biology** *Nat Rev Mol Cell Biol* **11**:849–60
3. Caussinus E, Gonzalez C (2005) **Induction of tumor growth by altered stem-cell asymmetric division in *Drosophila melanogaster*** *Nat Genet* **37**:1125–9
4. Gateff E (1978) **Malignant neoplasms of genetic origin in *Drosophila melanogaster*** *Science* **200**:1448–59
5. Albertson R, Dlg Doe CQ. (2003) **Scrib and Lgl regulate neuroblast cell size and mitotic spindle asymmetry** *Nat Cell Biol* **5**:166–70
6. Bello B, Reichert H, Hirth F (2006) **The brain tumor gene negatively regulates neural progenitor cell proliferation in the larval central brain of *Drosophila*** *Development* **133**:2639–48
7. Betschinger J, Mechtler K, Knoblich JA (2006) **Asymmetric segregation of the tumor suppressor brat regulates self-renewal in *Drosophila* neural stem cells** *Cell* **124**:1241–53
8. Bowman SK, Rolland V, Betschinger J, Kinsey KA, Emery G, Knoblich JA (2008) **The tumor suppressors Brat and Numb regulate transit-amplifying neuroblast lineages in *Drosophila*** *Dev Cell* **14**:535–46
9. Ohshiro T, Yagami T, Zhang C, Matsuzaki F (2000) **Role of cortical tumour-suppressor proteins in asymmetric division of *Drosophila* neuroblast** *Nature* **408**:593–6
10. Peng CY, Manning L, Albertson R, Doe CQ (2000) **The tumour-suppressor genes lgl and dlg regulate basal protein targeting in *Drosophila* neuroblasts** *Nature* **408**:596–600
11. Lee CY, Wilkinson BD, Siegrist SE, Wharton RP, Doe CQ (2006) **Brat is a Miranda cargo protein that promotes neuronal differentiation and inhibits neuroblast self-renewal** *Dev Cell* **10**:441–9
12. Bonnet D, Dick JE (1997) **Human acute myeloid leukemia is organized as a hierarchy that originates from a primitive hematopoietic cell** *Nat Med* **3**:730–7
13. Lapidot T, Sirard C, Vormoor J, Murdoch B, Hoang T, Caceres-Cortes J, et al. (1994) **A cell initiating human acute myeloid leukaemia after transplantation into SCID mice** *Nature* **367**:645–8
14. Bajaj J, Diaz E, Reya T (2020) **Stem cells in cancer initiation and progression** *J Cell Biol* :219
15. Reya T, Morrison SJ, Clarke MF, Weissman IL (2001) **Stem cells, cancer, and cancer stem cells** *Nature* **414**:105–11

16. Moore N, Lyle S (2011) **Quiescent, slow-cycling stem cell populations in cancer: a review of the evidence and discussion of significance** *J Oncol* **2011**
17. Rosen JM, Jordan CT (2009) **The increasing complexity of the cancer stem cell paradigm** *Science* **324**:1670–3
18. Yadav UP, Singh T, Kumar P, Sharma P, Kaur H, Sharma S, et al. (2020) **Metabolic Adaptations in Cancer Stem Cells** *Front Oncol* **10**:1010
19. Bajaj J, Zimdahl B, Reya T (2015) **Fearful symmetry: subversion of asymmetric division in cancer development and progression** *Cancer Res* **75**:792–7
20. Chao S, Yan H, Bu P (2024) **Asymmetric division of stem cells and its cancer relevance** *Cell Regen* **13**:5
21. Galli R, Binda E, Orfanelli U, Cipelletti B, Gritti A, De Vitis S, et al. (2004) **Isolation and characterization of tumorigenic, stem-like neural precursors from human glioblastoma** *Cancer Res* **64**:7011–21
22. Singh SK, Clarke ID, Terasaki M, Bonn VE, Hawkins C, Squire J, et al. (2003) **Identification of a cancer stem cell in human brain tumors** *Cancer Res* **63**:5821–8
23. Singh SK, Hawkins C, Clarke ID, Squire JA, Bayani J, Hide T, et al. (2004) **Identification of human brain tumour initiating cells** *Nature* **432**:396–401
24. Biserova K, Jakovlevs A, Uljanovs R, Strumfa I (2021) **Cancer Stem Cells: Significance in Origin, Pathogenesis and Treatment of Glioblastoma** *Cells* :10
25. Chen J, Li Y, Yu TS, McKay RM, Burns DK, Kernie SG, et al. (2012) **A restricted cell population propagates glioblastoma growth after chemotherapy** *Nature* **488**:522–6
26. Mesrati M Hassn, Behrooz AB, Abuhamad A Y, Syahir A (2020) **Understanding Glioblastoma Biomarkers: Knocking a Mountain with a Hammer** *Cells* :9
27. Tang X, Zuo C, Fang P, Liu G, Qiu Y, Huang Y, et al. (2021) **Targeting Glioblastoma Stem Cells: A Review on Biomarkers, Signal Pathways and Targeted Therapy** *Front Oncol* **11**:701291
28. Larriba E, de Juan Romero C, García-Martínez A, Quintanar T, Rodríguez- Lescure Á, Soto JL, et al. (2024) **Identification of new targets for glioblastoma therapy based on a DNA expression microarray** *Comput Biol Med* **179**:108833
29. Chen G, Kong J, Tucker-Burden C, Anand M, Rong Y, Rahman F, et al. (2014) **Human Brat ortholog TRIM3 is a tumor suppressor that regulates asymmetric cell division in glioblastoma** *Cancer Res* **74**:4536–48
30. Wang L, Zhan W, Xie S, Hu J, Shi Q, Zhou X, et al. (2014) **Over-expression of Rap2a inhibits glioma migration and invasion by down-regulating p-AKT** *Cell Biol Int* **38**:326–34
31. Wang L, Zhu B, Wang S, Wu Y, Zhan W, Xie S, et al. (2017) **Regulation of glioma migration and invasion via modification of Rap2a activity by the ubiquitin ligase Nedd4-1** *Oncol Rep* **37**:2565–74
32. Carmena A, Makarova A, Speicher S (2011) **The Rap1-Rgl-Ral signaling network regulates neuroblast cortical polarity and spindle orientation** *J Cell Biol* **195**:553–62

33. Bello BC, Izergina N, Caussinus E, Reichert H (2008) **Amplification of neural stem cell proliferation by intermediate progenitor cells in Drosophila brain development** *Neural Dev* **3**:5
34. Boone JQ, Doe CQ (2008) **Identification of Drosophila type II neuroblast lineages containing transit amplifying ganglion mother cells** *Dev Neurobiol* **68**:1185–95
35. Wodarz A, Ramrath A, Grimm A, Knust E (2000) **Drosophila atypical protein kinase C associates with Bazooka and controls polarity of epithelia and neuroblasts** *J Cell Biol* **150**:1361–74
36. Knoblich JA, Jan LY, Jan YN (1995) **Asymmetric segregation of Numb and Prospero during cell division** *Nature* **377**:624–7
37. Speicher S, Fischer A, Knoblich J, Carmena A (2008) **The PDZ protein Canoe regulates the asymmetric division of Drosophila neuroblasts and muscle progenitors** *Curr Biol* **18**:831–7
38. Gerdes J, Schwab U, Lemke H, Stein H (1983) **Production of a mouse monoclonal antibody reactive with a human nuclear antigen associated with cell proliferation** *Int J Cancer* **31**:13–20
39. Sun X, Kaufman PD (2018) **Ki-67: more than a proliferation marker** *Chromosoma* **127**:175–86
40. Yan B (2010) **Numb--from flies to humans** *Brain Dev* **32**:293–8
41. Morrison SJ, Kimble J (2006) **Asymmetric and symmetric stem-cell divisions in development and cancer** *Nature* **441**:1068–74
42. Cayouette M, Raff M (2002) **Asymmetric segregation of Numb: a mechanism for neural specification from Drosophila to mammals** *Nat Neurosci* **5**:1265–9
43. Cicalese A, Bonizzi G, Pasi CE, Faretta M, Ronzoni S, Giulini B, et al. (2009) **The tumor suppressor p53 regulates polarity of self-renewing divisions in mammary stem cells** *Cell* **138**:1083–95
44. Ortega-Campos SM, García-Heredia JM (2023) **The Multitasker Protein: A Look at the Multiple Capabilities of NUMB** *Cells* :12
45. Li Z, Zhang YY, Zhang H, Yang J, Chen Y, Lu H (2022) **Asymmetric Cell Division and Tumor Heterogeneity** *Front Cell Dev Biol* **10**:938685
46. Dey-Guha I, Wolfer A, Yeh AC, Albeck J G, Darp R, Leon E, et al. (2011) **Asymmetric cancer cell division regulated by AKT** *Proc Natl Acad Sci U S A* **108**:12845–50
47. Sugiarto S, Persson AI, Munoz EG, Waldhuber M, Lamagna C, Andor N, et al. (2011) **Asymmetry-defective oligodendrocyte progenitors are glioma precursors** *Cancer Cell* **20**:328–40
48. Daynac M, Chouchane M, Collins HY, Murphy NE, Andor N, Niu J, et al. (2018) **Lgl1 controls NG2 endocytic pathway to regulate oligodendrocyte differentiation and asymmetric cell division and gliomagenesis** *Nat Commun* **9**:2862
49. Bu P, Chen KY, Chen JH, Wang L, Walters J, Shin YJ, et al. (2013) **A microRNA miR-34a-regulated bimodal switch targets Notch in colon cancer stem cells** *Cell Stem Cell* **12**:602–15

50. Hwang WL, Jiang JK, Yang SH, Huang TS, Lan HY, Teng HW, et al. (2014) **MicroRNA- 146a directs the symmetric division of Snail-dominant colorectal cancer stem cells** *Nat Cell Biol* **16**:268–80
51. Homem CC, Knoblich JA (2012) **Drosophila neuroblasts: a model for stem cell biology** *Development* **139**:4297–310
52. Samanta P, Bhowmik A, Biswas S, Sarkar R, Ghosh R, Pakhira S, et al. (2023) **Therapeutic Effectiveness of Anticancer Agents Targeting Different Signaling Molecules Involved in Asymmetric Division of Cancer Stem Cell** *Stem Cell Rev Rep* **19**:1283–306
53. Chao S, Zhang F, Yan H, Wang L, Zhang L, Wang Z, et al. (2023) **Targeting intratumor heterogeneity suppresses colorectal cancer chemoresistance and metastasis** *EMBO Rep* **24**:e56416
54. Hitomi M, Chumakova AP, Silver DJ, Knudsen AM, Pontius WD, Murphy S, et al. (2021) **Asymmetric cell division promotes therapeutic resistance in glioblastoma stem cells** *JCI Insight* **6**
55. de Thé H (2018) **Differentiation therapy revisited** *Nat Rev Cancer* **18**:117–27
56. Neumuller RA, Richter C, Fischer A, Novatchkova M, Neumuller KG, Knoblich JA (2011) **Genome-wide analysis of self-renewal in Drosophila neural stem cells by transgenic RNAi** *Cell Stem Cell* **8**:580–93
57. Rives-Quinto N, Franco M, de Torres-Jurado A, Carmena A (2017) **Synergism between canoe and scribble mutations causes tumor-like overgrowth via Ras activation in neural stem cells and epithelia** *Development* **144**:2570–83
58. Saeed AI, Sharov V, White J, Li J, Liang W, Bhagabati N, et al. (2003) **TM4: a free, open- source system for microarray data management and analysis** *Biotechniques* **34**:374–8
59. Gargini R, Escoll M, García E, García-Escudero R, Wandosell F, Antón IM (2016) **WIP Drives Tumor Progression through YAP/TAZ-Dependent Autonomous Cell Growth** *Cell Rep* **17**:1962–77
60. Gargini R, Segura-Collar B, Herránz B, García-Escudero V, Romero-Bravo A, Núñez FJ, et al. (2020) **The IDH-TAU-EGFR triad defines the neovascular landscape of diffuse gliomas** *Sci Transl Med* **12**
61. Ponti D, Costa A, Zaffaroni N, Pratesi G, Petrangolini G, Coradini D, et al. (2005) **Isolation and in vitro propagation of tumorigenic breast cancer cells with stem/progenitor cell properties** *Cancer Res* **65**:5506–11
62. Gargini R, Cerliani JP, Escoll M, Antón IM, Wandosell F (2015) **Cancer stem cell-like phenotype and survival are coordinately regulated by Akt/FoxO/Bim pathway** *Stem Cells* **33**:646–60

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Reviewer #1 (Public review):

Summary:

The authors validate the contribution of RAP2A to GB progression. RAP2A participates in asymmetric cell division, and the localization of several cell polarity markers, including cno and Numb.

Strengths:

The use of human data, Drosophila models, and cell culture or neurospheres is a good scenario to validate the hypothesis using complementary systems.

Moreover, the mechanisms that determine GB progression, and in particular glioma stem cells biology, are relevant for the knowledge on glioblastoma and opens new possibilities to future clinical strategies.

Weaknesses:

While the manuscript presents a well-supported investigation into RAP2A's role in GBM, several methodological aspects require further validation. The major concern is the reliance on a single GB cell line (GB5), which limits the generalizability of the findings. Including multiple GBM lines, particularly primary patient-derived 3D cultures with known stem-like properties, would significantly enhance the study's relevance.

Additionally, key mechanistic aspects remain underexplored. Further investigation into the conservation of the Rap2l-Cno/aPKC pathway in human cells through rescue experiments or protein interaction assays would be beneficial. Similarly, live imaging or lineage tracing would provide more direct evidence of ACD frequency, complementing the current indirect metrics (odd/even cell clusters, Numb asymmetry).

Several specific points raised in previous reviews still require attention:

- (1) The specificity of Rap2l RNAi needs further confirmation. Is Rap2l expressed in neuroblasts or intermediate neural progenitors? Can alternative validation methods be employed?
- (2) Quantification of phenotypic penetrance and survival rates in Rap2l mutants would help determine the consistency of ACD defects.
- (3) The observations on neurosphere size and Ki-67 expression require normalization (e.g., Ki-67+ cells per total cell number or per neurosphere size). Additionally, apoptosis should be assessed using Annexin V or TUNEL assays.
- (4) The discrepancy in Figures 6A and 6B requires further discussion.
- (5) Live imaging of ACD events would provide more direct evidence.
- (6) Clarification of terminology and statistical markers (e.g., p-values) in Figure 1A would improve clarity.
- (7) Given the group's expertise, an alternative to mouse xenografts could be a *Drosophila* genetic model of glioblastoma, which would provide an *in vivo* validation system aligned with their research approach.

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Reviewer #2 (Public review):

This study investigates the role of RAP2A in regulating asymmetric cell division (ACD) in glioblastoma stem cells (GSCs), bridging insights from *Drosophila* ACD mechanisms to human tumor biology. They focus on RAP2A, a human homolog of *Drosophila* Rap2l, as a novel ACD regulator in GBM is innovative, given its underexplored role in cancer stem cells (CSCs). The hypothesis that ACD imbalance (favoring symmetric divisions) drives GSC expansion and tumor progression introduces a fresh perspective on differentiation therapy. However, the dual role of ACD in tumor heterogeneity (potentially aiding therapy resistance) requires deeper discussion to clarify the study's unique contributions against existing controversies. Some limitations and questions need to be addressed.

- (1) Validation of RAP2A's prognostic relevance using TCGA and Gravendeel cohorts strengthens clinical relevance. However, differential expression analysis across GBM subtypes (e.g., MES, DNA-methylation subtypes) should be included to confirm specificity.
- (2) Rap2l knockdown-induced ACD defects (e.g., mislocalization of Cno/Numb) are well-designed. However, phenotypic penetrance and survival rates of Rap2l mutants should be

quantified to confirm consistency.

(3) While GB5 cells were effectively used, justification for selecting this line (e.g., representativeness of GBM heterogeneity) is needed. Experiments in additional GBM lines (especially the addition of 3D primary patient-derived cell lines with known stem cell phenotype) would enhance generalizability.

(4) Indirect metrics (odd/even cell clusters, NUMB asymmetry) are suggestive but insufficient. Live imaging or lineage tracing would directly validate ACD frequency.

(5) The initial microarray (n=7 GBM patients) is underpowered. While TCGA data mitigate this, the limitations of small cohorts should be explicitly addressed and need to be discussed.

(6) Conclusions rely heavily on neurosphere models. Xenograft experiments or patient-derived orthotopic models are critical to support translational relevance, and such basic research work needs to be included in journals.

(7) How does RAP2A regulate NUMB asymmetry? Is the *Drosophila* Rap2l-Cno/aPKC pathway conserved? Rescue experiments (e.g., Cno/aPKC knockdown with RAP2A overexpression) or interaction assays (e.g., Co-IP) are needed to establish molecular mechanisms.

(8) Reduced stemness markers (CD133/SOX2/NESTIN) and proliferation (Ki-67) align with increased ACD. However, alternative explanations (e.g., differentiation or apoptosis) must be ruled out via GFAP/Tuj1 staining or Annexin V assays.

(9) The link between low RAP2A and poor prognosis should be validated in multivariate analyses to exclude confounding factors (e.g., age, treatment history).

(10) The broader ACD regulatory network in GBM (e.g., roles of other homologs like NUMB) and potential synergies/independence from known suppressors (e.g., TRIM3) warrant exploration.

(11) The figures should be improved. Statistical significance markers (e.g., p-values) should be added to Figure 1A; timepoints/culture conditions should be clarified for Figure 6A.

(12) Redundant *Drosophila* background in the Discussion should be condensed; terminology should be unified (e.g., "neurosphere" vs. "cell cluster").

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